SCHOOL OF COMPUTER SCIENCE
GRADUATION PROJECT DOCUMENTATION
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InsightX

Submitted by

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Abstract

In today's fast-paced e-commerce landscape, data has evolved into the most powerful asset for driving strategic growth. Yet a hidden crisis exists; small-to-medium enterprise owners are drowning in raw data but starving for actionable insights. Without enterprise-grade analytical tools, these businesses are forced to rely on intuition to forecast inventory, understand customer sentiment, and identify VIP buyers.

InsightX was engineered to solve this crisis. Acting as an automated "Data Scientist in a Box," InsightX is a comprehensive SaaS platform that democratizes Artificial Intelligence. By integrating a cloud-native microservices architecture with advanced Machine Learning—specifically Facebook Prophet for time-series forecasting and a Multilingual BERT Transformer for language-agnostic sentiment analysis—the platform transforms chaotic, real-world data into predictive intelligence. InsightX empowers business owners to stop guessing and start forecasting, delivering real-time strategic foresight through a highly interactive, zero-code dashboard.

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List of acronyms/abbreviations

Abbreviations	Word
CIC	Canadian International College
PC	Personal Computer
SMEs	small-to-medium enterprises

CHAPTER 1: INTRODUCTION

1.1 Background and overview

In today's fast-paced e-commerce landscape, data has evolved into the most powerful asset for driving strategic growth. Imagine a local entrepreneur who just launched a high-quality e-commerce brand. They have a great product and just hit a record-breaking sales month. However, when asked *Why* a specific item sold out, or *exactly how much* inventory they need to order for the next quarter, they freeze. They open their laptop and are greeted by thousands of rows of raw data: spreadsheets full of customer reviews, order timestamps, and complex inventory logs. Without enterprise-grade analytical tools, these businesses are forced to rely on intuition rather than proper data analysis to forecast inventory and understand customer sentiment.

1.2 Problem statement

Most businesses face significant challenges when managing and interpreting their data. While enterprise giants use armies of data scientists to predict market trends, small-to-medium enterprises are left in the dark. Existing tools like Tableau or Power BI are incredibly powerful but require a steep technical learning curve to use effectively. On the other end, built-in dashboards on platforms like Shopify only show *what happened yesterday*, offering zero predictive power for *what will happen tomorrow*. The problem is no longer a lack of data; the problem is the inability to translate that data into a forward-looking strategy. This lack of insight leads to inaccurate forecasts, a poor understanding of customer needs, and an inability to plan for future growth.

1.3 Project objectives

The main objective of InsightX is to act as an automated "Data Scientist in a Box," developing an intelligent and user-friendly web application that empowers

businesses to utilize their data effectively. The specific objectives of this project are:

- **Predictive Analytics:** To develop Machine Learning models (e.g., Facebook Prophet) capable of forecasting future sales and market demand.
- **Prescriptive Analytics:** To implement Natural Language Processing (BERT) to analyze the sentiment of customer reviews instantly.
- **Descriptive Analytics:** To automatically cluster customers into behavioral segments, identifying VIPs versus those at risk of churning.
- **Automation:** To automate the process of data cleaning and processing, ensuring that uploaded datasets are ready for visualization without manual intervention.

1.4 Project scope

The scope of this project is defined by the development of a cloud-native analytics platform tailored to the e-commerce sector.

- **In Scope:** The system will act as an all-in-one platform integrating data ingestion, automated cleaning, and interactive visualization via a web dashboard. It will feature a **Language-Agnostic NLP Engine** that processes native product reviews in over 100 languages without reliance on external translation of APIs, preserving cultural nuance.
- **Out of Scope:** The project does not include the development of the e-commerce store itself (e.g., shopping cart or payment processing) but rather focuses strictly on the analytics and forecasting layer that processes the store's data.

1.5 Contributions/Significance of the Project

The significance of InsightX lies in its ability to democratize Artificial Intelligence for SMEs. By bridging the gap between raw data and predictive AI, InsightX allows business owners to stop reacting to the past and start engineering their

future. It provides a cost-effective, automated solution that eliminates the need for expensive data science teams, transforming chaotic, real-world data into predictive intelligence with zero coding required from the user.

1.6 Project organization

The remainder of this documentation is organized as follows: Chapter 2 explores related works and market gaps. Chapter 3 details the system analysis and design, including UML diagrams and the cloud-native architecture. Chapter 4 describes the implementation phase, including the technology stack and database design. Finally, Chapter 5 summarizes the project's achievements and outlines future work.

1.7 Conclusion

This chapter introduced InsightX as a vital solution for modern e-commerce businesses struggling with data management. We defined the core problem of intuition-based decision-making and outlined our objective to build an intelligent, automated forecasting platform

CHAPTER 2: INVESTIGATIONS

2.1 Overview

This chapter explores modern, AI-driven solutions in the field of e-commerce analytics. When an SME owner searches for software, they find a market broken into extremes. We investigated current market leaders focusing on predictive analytics and automated machine learning (AutoML) to identify the specific gaps that InsightX will address.

2.2 Summary of related works

Triple Whale is a modern e-commerce operating system widely used by direct-to-consumer (DTC) brands. It centralizes metrics from storefronts and ad platforms.

- **Strengths:** Outstanding user interface and real-time data aggregation. It excels at tracking Return on Ad Spend (ROAS).

- **Limitations:** It is heavily focused on marketing attribution rather than deep inventory forecasting or natural language sentiment analysis.

Klaviyo (Predictive Analytics Module) Klaviyo is an advanced marketing automation platform that includes a robust predictive analytics engine for calculating churn risk.

- **Strengths:** Highly accurate customer segmentation and automated email triggers based on predictive data.

- **Limitations:** The predictive tools are locked within an email marketing ecosystem. It does not provide general sales forecasting for the business layer.

DataRobot (Enterprise AutoML) DataRobot is an enterprise AI platform that automates the process of building and deploying machine learning models.

- **Strengths:** Extremely powerful. It can test hundreds of algorithms simultaneously to find the most accurate model.

- **Limitations:** It is an infrastructure tool designed for technical data science teams, making it entirely inaccessible and too expensive for the SME market.

2.3 Comparison between related works

Feature	**Triple Whale**	**Klaviyo**	**DataRobot**	**InsightX (Proposed)**
Primary Focus	Marketing ROI	Email Automation	Enterprise ML	**E-commerce Forecasting**
Target Audience	Mid-Market Brands	Marketers	Data Scientists	**SME Business Owners**

Sales Forecasting	Limited	No	Yes (Manual build)	**Yes (Automated Prophet)**
NLP / Sentiment	No	No	Yes (Manual build)	**Yes (Automated BERT)**
Ease of Use	High	High	Low (Technical)	**High (Upload & Analyze)**

2.4 Conclusion

While the e-commerce analytics market has evolved, a distinct gap remains. Platforms like Triple Whale excel in marketing but lack broad forecasting capabilities. DataRobot offers immense AI power but requires a highly technical team. InsightX bridges this gap by offering a streamlined, "AI-in-a-Box" solution that brings enterprise-grade forecasting and sentiment analysis directly to SMEs without the steep learning curve.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter details the "Engine Room" of InsightX. To deliver a seamless experience to the end-user, the underlying architecture must handle massive computational complexity invisibly. This section outlines our Agile methodology, the functional requirements, and the sophisticated architectural design required to process heavy machine learning tasks in real-time.

3.2 Methodology

For the development of InsightX, we selected Agile Methodology, specifically the Scrum framework. Developing Machine Learning models is highly experimental. This iterative approach allowed seamless, continuous integration between the Data Science team (building the predictive models) and the Cloud Computing

team (deploying the infrastructure), ensuring that AI accuracy improved with every sprint.

3.3 System Analysis

3.3.1 Functional and Non-functional Requirements

- **FR01 - User Authentication:** The system must allow users to register and log in securely using encrypted credentials (JWT/OAuth2).
- **FR02 - Data Upload:** The system must allow users to upload CSV or Excel files containing raw sales, inventory, and customer review data.
- **FR03 - Data Processing:** The system must automatically validate and cleanupload data, handling missing values, removing duplicates, and standardizing date formats.
- **FR04 - Entity Parsing:** The system must utilize Natural Language Processing (NLP) to extract specific entities (e.g., Product Names, Colors, Brands) from unstructured text columns within the dataset.
- **FR05 - Sales Forecasting:** The system must utilize Time-Series Machine Learning (e.g., Prophet) to predict future sales volumes based on historical transaction timestamps.
- **FR06 - Customer Segmentation:** The system must apply clustering algorithms to categorize customers into behavioral groups (e.g., "Big Spenders," "One-Time Buyers").
- **FR07 - Sentiment Analysis:** The system must analyze the text of customer reviews to calculate a sentiment score (Positive, Negative, Neutral) and identify quality issues.
- **FR08 - Dashboard Visualization:** The system must generate interactive visualizations (Line charts for forecasts, Scatter plots for segmentation) to display Key Performance Indicators (KPIs).
- **FR09 - Dashboard Customization:** The system must provide a flexible interface allowing users to resize, rearrange, and toggle the visibility of specific widgets to personalize their view.
- **FR10 - Business Scenario Simulation:** The system must provide an interactive "What-If" simulation tool, allowing users to adjust variables (e.g., "Increase Price by 10%") and instantly visualize the predicted impact on revenue.

Non-Functional Requirements (How the system behaves):

- **NFR01 - Performance:** The dashboard should load visualization results within 3 seconds of a request.
- **NFR02 - Scalability:** The system architecture must support multiple concurrent users without degradation in speed.
- **NFR03 - Reliability:** The forecasting service must be available 99% of the time.
- **NFR04 - Usability:** The user interface must be intuitive enough for non-technical business owners to navigate without training.

3.3.2 Software and Hardware Requirements

A. Software Requirements: To ensure scalability and performance for the InsightX platform, the following technology stack has been selected:

- **Frontend Development:**
    - **Framework:** React.js or Next.js (for building a dynamic, responsive user interface).
    - **Styling:** Tailwind CSS or Material UI.
- **Backend & API:**
    - **Language:** Python 3.10+ (Selected for its native support of Data Science libraries).
    - **Framework:** Fast API or Django REST Framework (for handling high-performance API requests).
- **Data Science & Machine Learning:**
    - **Libraries:** Pandas (data manipulation), NumPy (numerical analysis), Scikit-Learn (predictive modeling).
    - **Forecasting Engine:** Facebook Prophet or ARIMA models.
- **Database Management:**
    - **Primary Database:** PostgreSQL (for structured relational data like sales and user profiles).

    - **Data Storage:** AWS S3 or Google Cloud Storage (for storing large raw CSV/Excel user uploads).

- **DevOps & Tools:**
    - **Containerization:** Docker (to ensure consistent environments between development and production.
    - **Version Control:** Git & GitHub.

B. Hardware Requirements:

- **1. Development Environment (Local Machines):**
    - **Processor:** Intel Core i5/i7 (10th Gen or later) or AMD Ryzen 5/7.
    - **RAM:** Minimum 16GB (Critical for Machine Learning models locally).
    - **Storage:** 512GB SSD (for fast data retrieval).
    - **OS:** Windows 10/11 or Linux (Ubuntu).
- **2. Deployment Environment (Cloud Infrastructure):**
    - **Cloud Provider:** Amazon Web Services) or Microsoft Azure.
    - **Compute:** AWS EC2 (t3. medium or larger) for hosting the application.
    - **Storage:** Scalable object storage (S3) for handling user datasets.
    - **Database Service:** Managed RDS (Relational Database Service) for high availability.

3.4 System Design

The system is built using a **3-tier architecture** to keep the platform organized and fast:

1. **Frontend (The Look):** The web dashboard where users see their charts and KPIs.
2. **Backend (The Brain):** The server that handles data cleaning and runs the predictive models.

3. **Database (The Storage):** A secure place to store user accounts and business information.

3.4.1 UML diagrams

3.4.1.1-System architecture diagram

1. **System Architecture Diagram:** InsightX is built on a Cloud-Native Microservices Architecture. The Public Subnet acts as a secure entry point via a Nginx Load Balancer. The Private Subnet hosts the Fast API Backend and our AutoML Workers. We utilize Polyglot Persistence—storing structured user metadata in PostgreSQL, while dumping unstructured CSV files into an AWS S3 Data Lake.

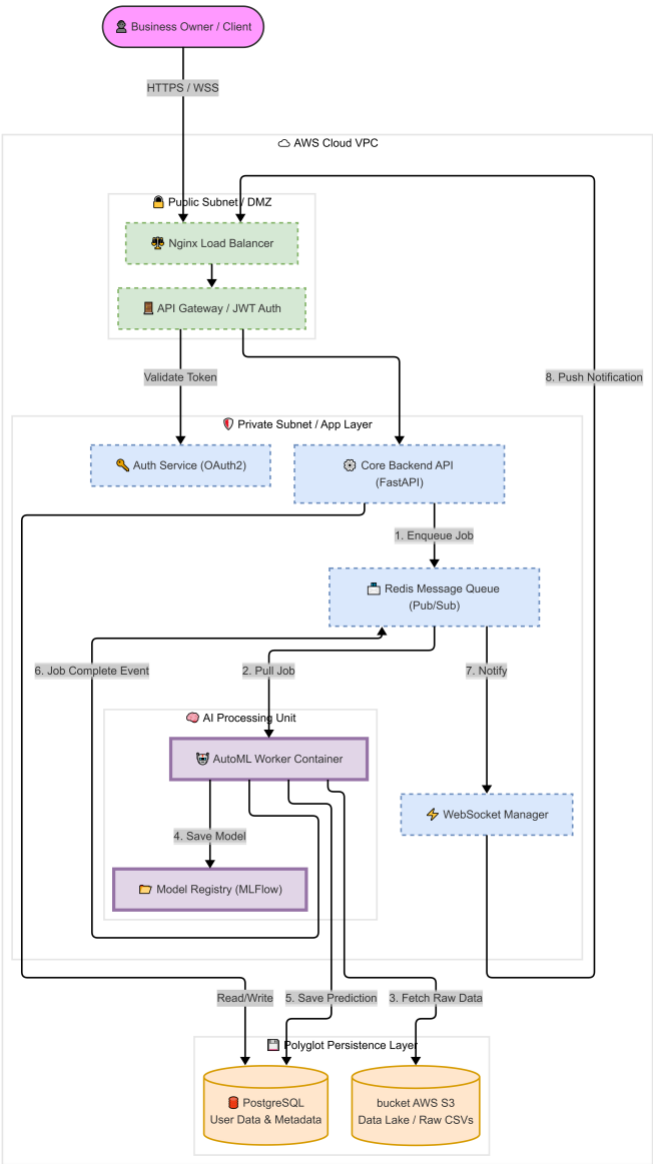
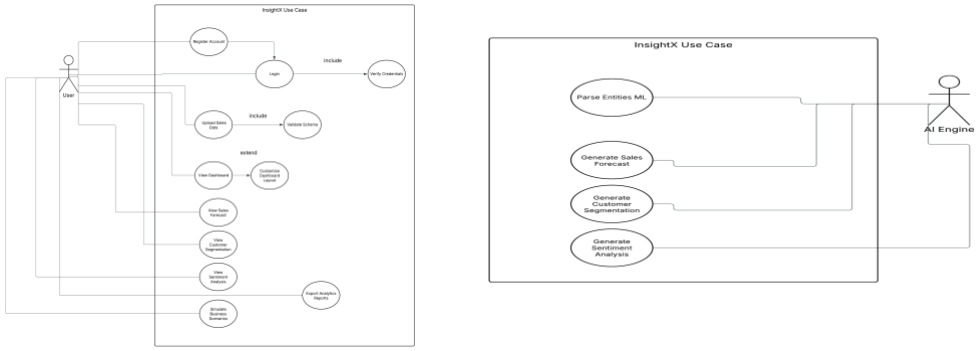


Figure 1: System architecture diagram

3.4.1.2- Use case diagram

Use Case Diagram: The Business Owner simply uploads data and views of insights. However, we explicitly model the **AI Engine** as a distinct System Actor. This highlights that the AI is an active computational participant that autonomously generates forecasts in the background.



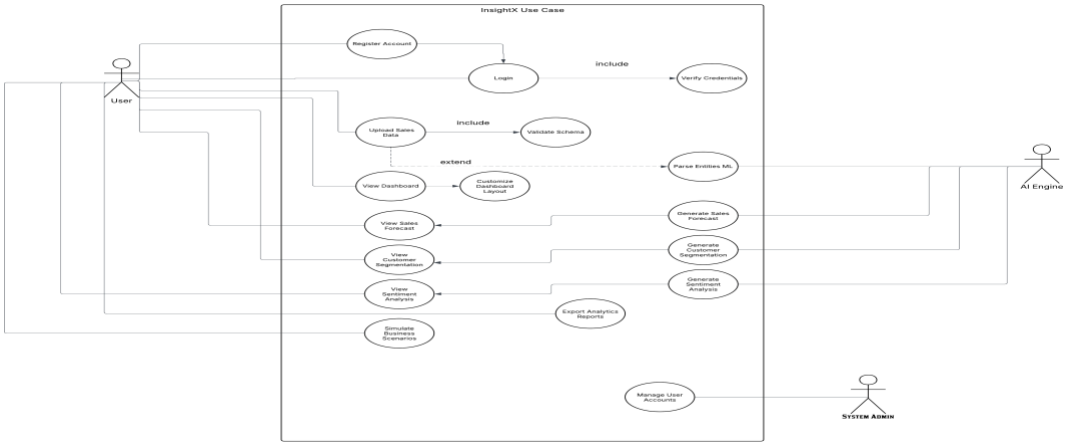


Figure 2: Use case diagram

3.4.1.3-Activity diagram

Activity Diagram: We employ a "Fail-Fast" validation protocol. Upon successful file validation, the system performs an Asynchronous Handoff. It pushes a job to Redis and immediately returns a "202 Accepted" status, ensuring the frontend never freezes while the AI worker cleans the data.

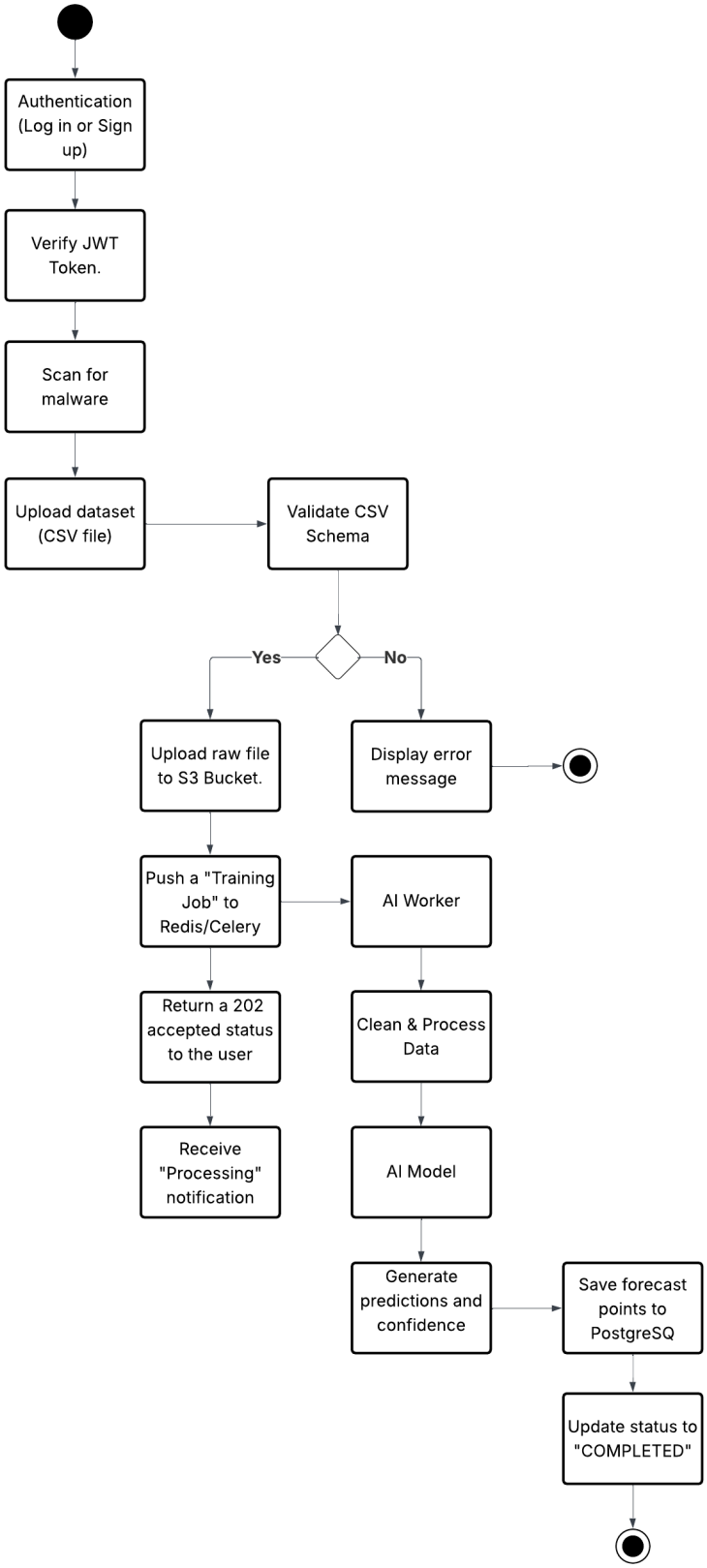


Figure 3: Activity diagram

3.4.1.4- Sequence diagram

Sequence Diagram: Training a Prophet forecasting model takes time. This diagram showcases our **Asynchronous Request-Reply Pattern**. By introducing a Redis Message Queue, we temporally decouple the API from the heavy lifting. The client simply polls the server for a status update, ensuring a responsive UI.

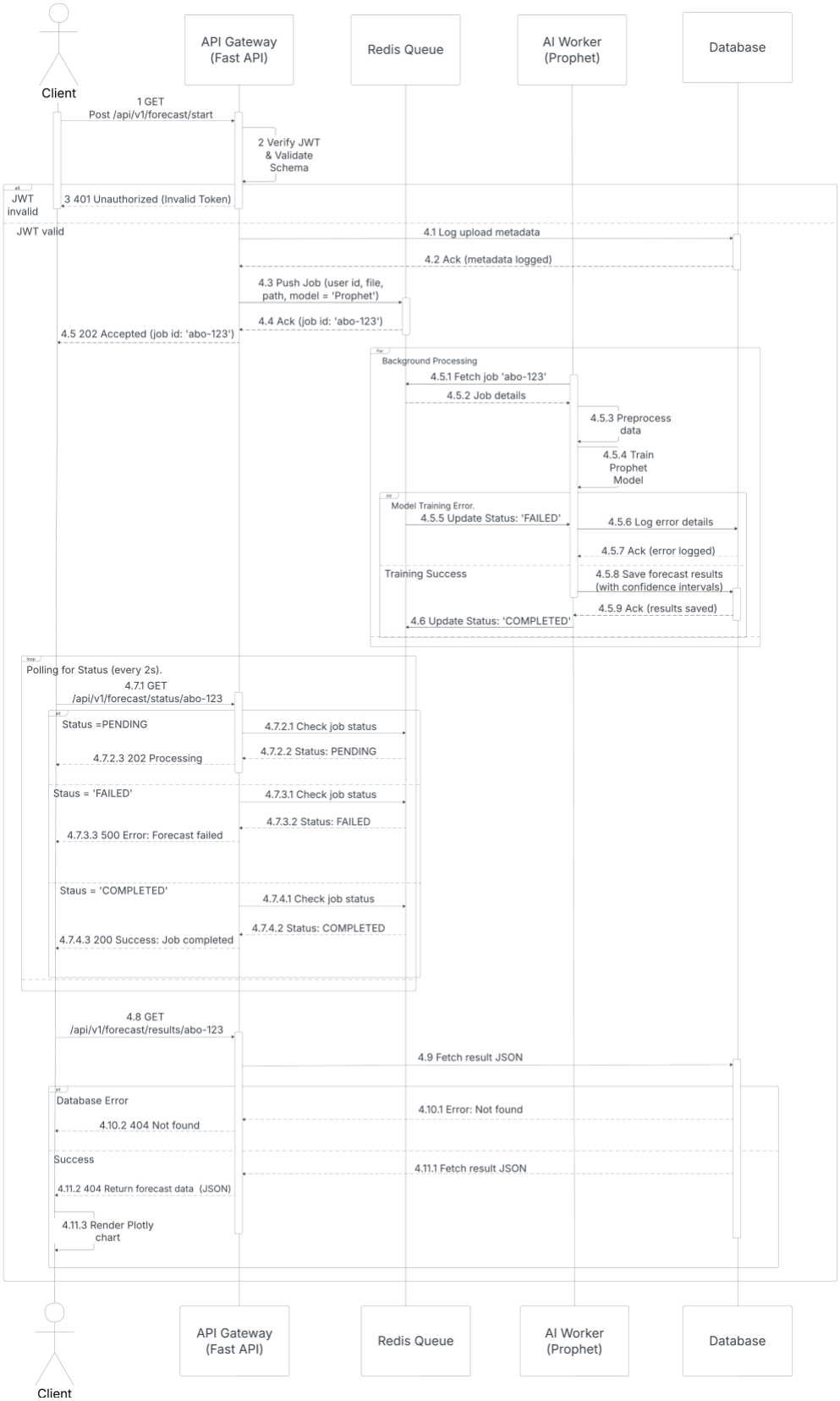


Figure 4: Sequence diagram

3.4.1.5- Class diagram

Class Diagram: We implemented the **Strategy Design Pattern**. By creating an `IForecastStrategy` interface, our system can dynamically swap between a Facebook Prophet model (for highly seasonal data) and an ARIMA model (for linear trends) without requiring changes to the core controller logic.

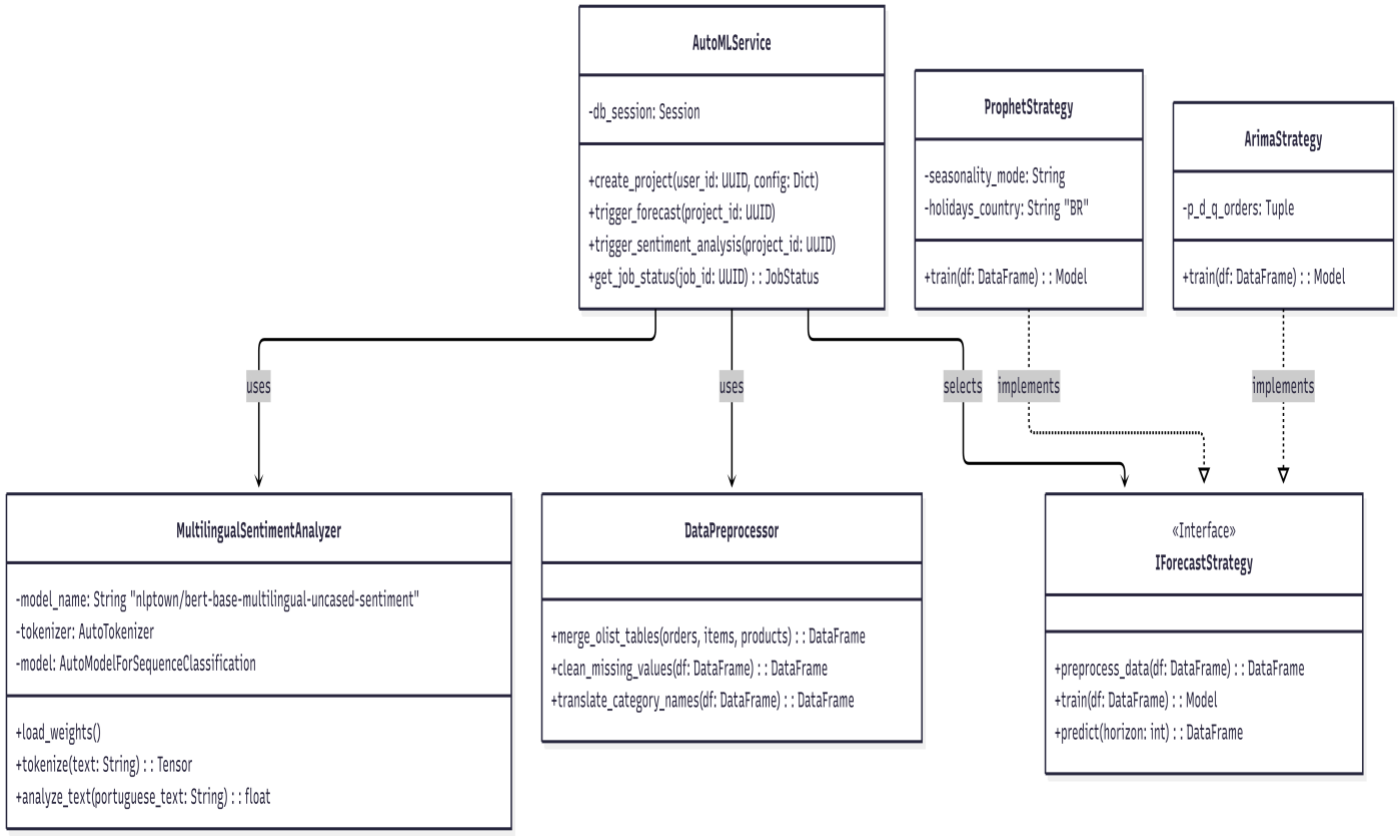


Figure 5: Class diagram

3.4.1.6 - ER Diagram

The database schema follows Third Normal Form (3NF). To guarantee that our dashboard loads in under 3 seconds, we engineered a dedicated **AI Results Layer** (`FORECAST_RESULT`, `SEGMENTATION_RESULT`, `SENTIMENT_RESULT`). The dashboard instantly fetches pre-computed insights from this layer rather than re-running algorithms.

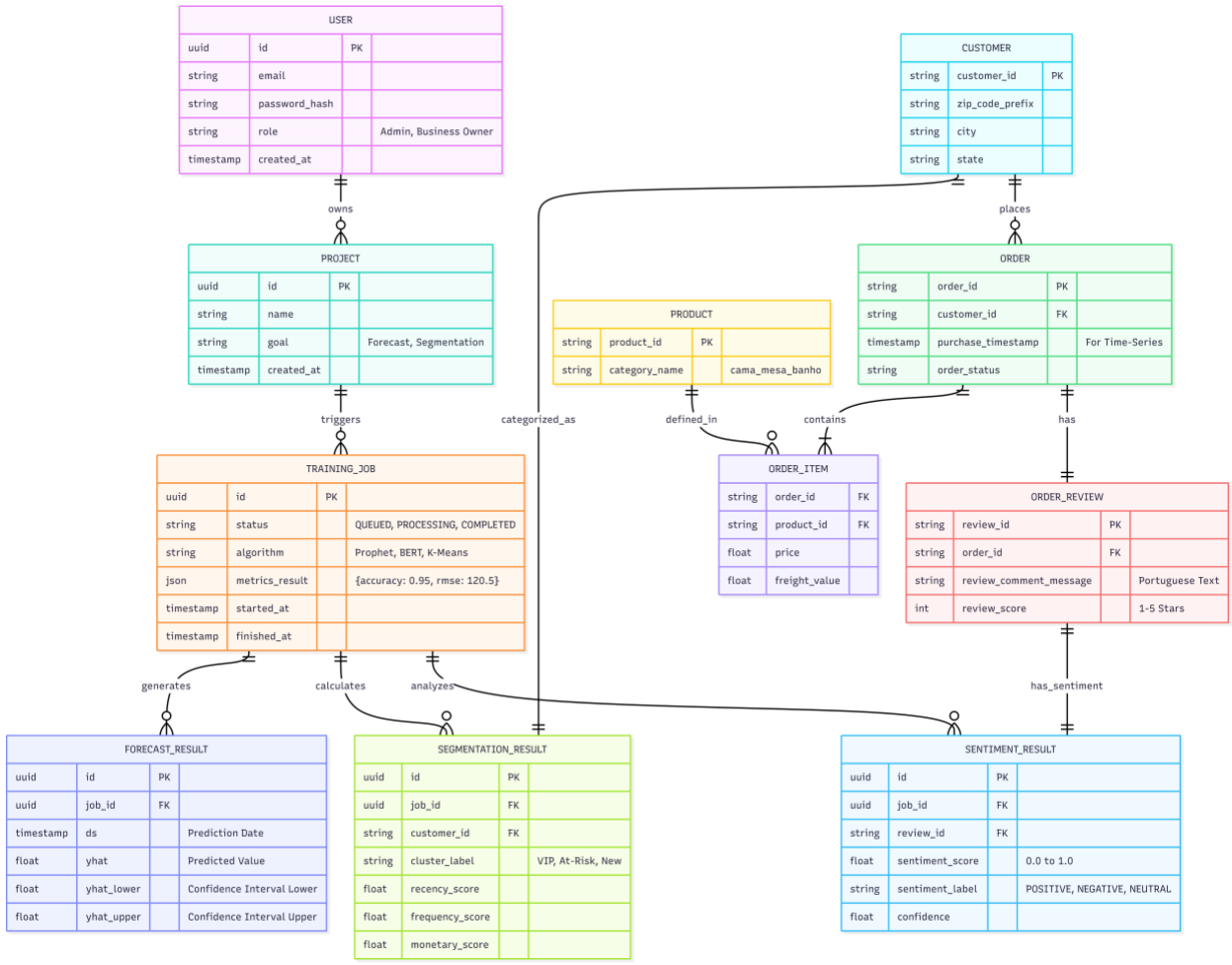
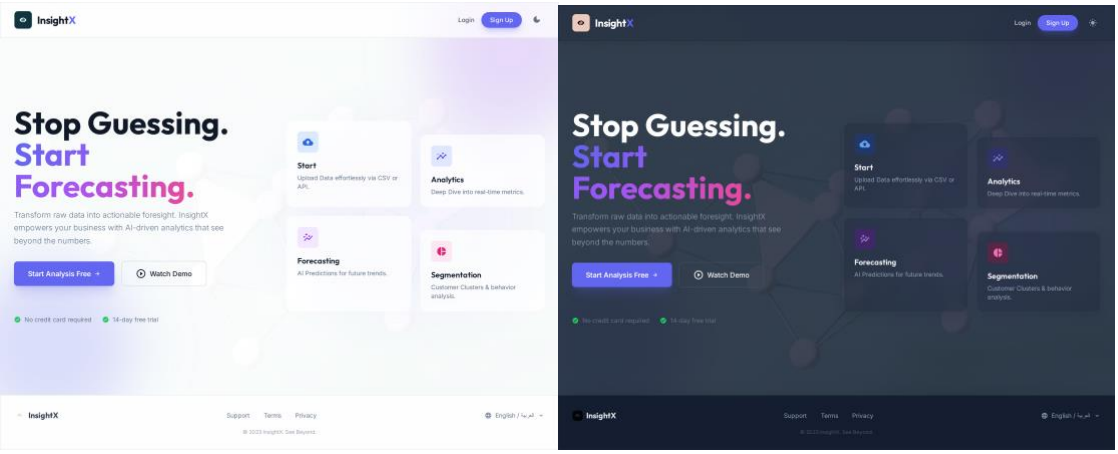


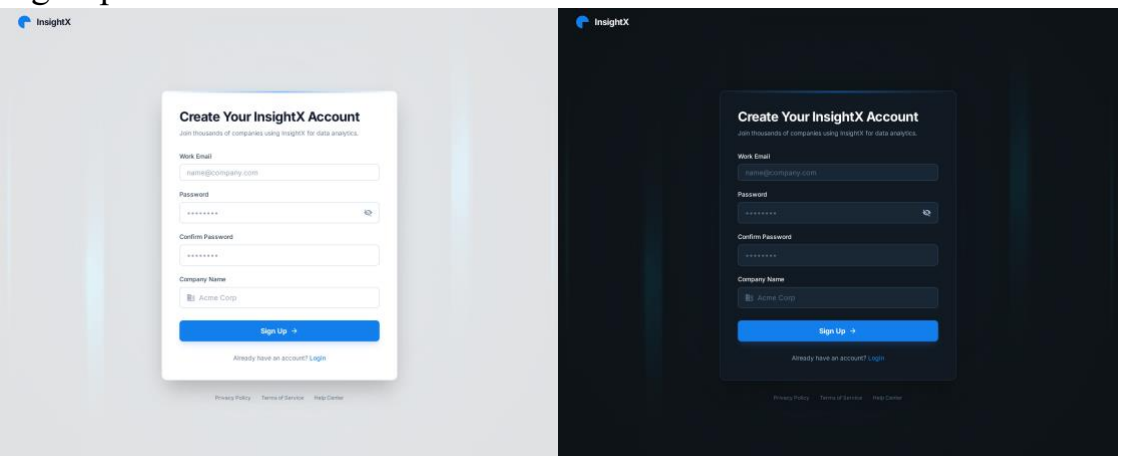
Figure 6: **ER Diagram**

3.4.2 User interface design

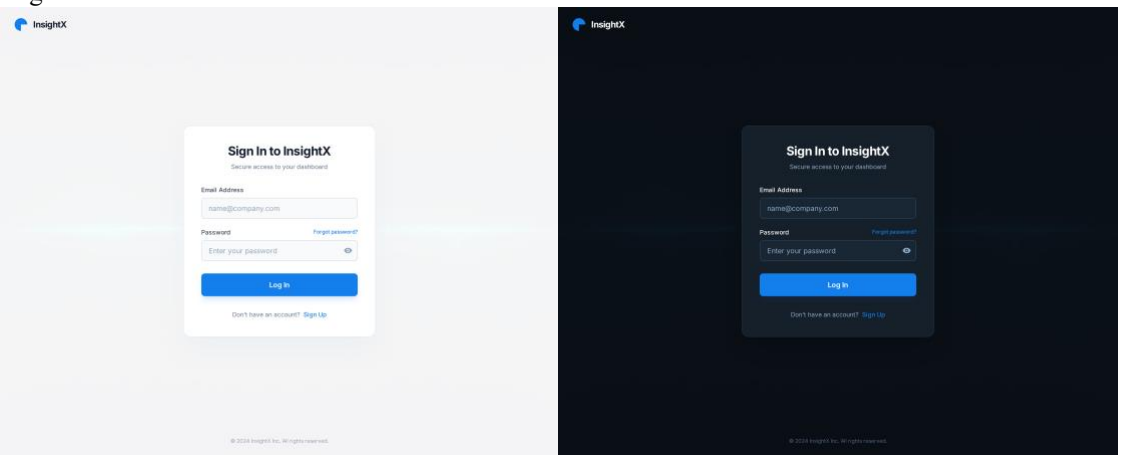
Landing page:



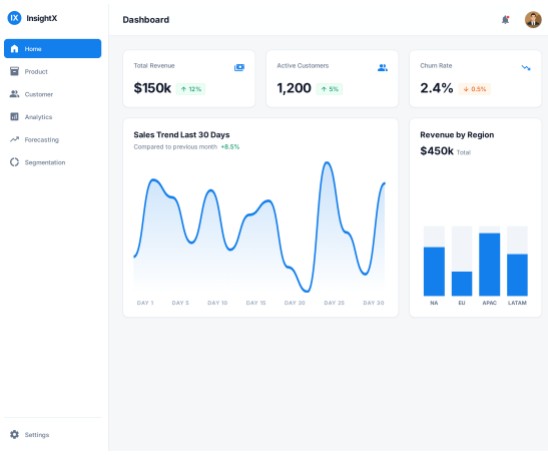
Sign-up :



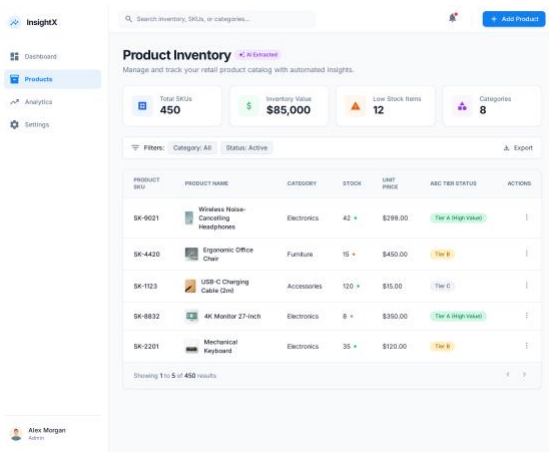
Login:

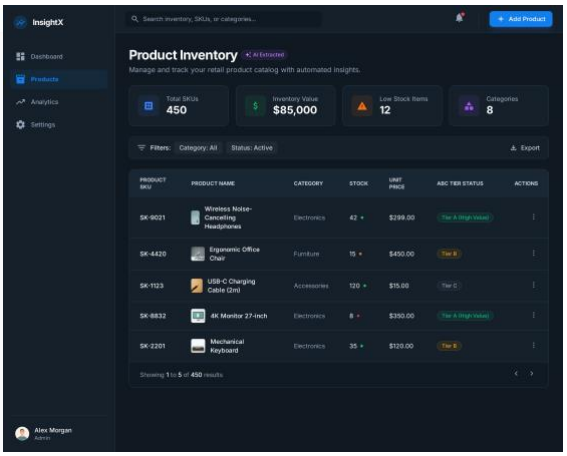


Dashboard Home :

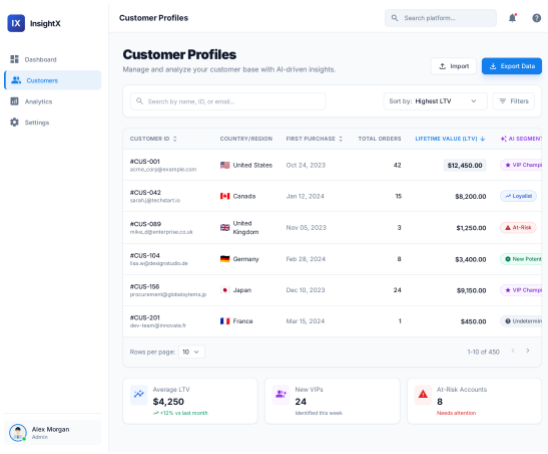


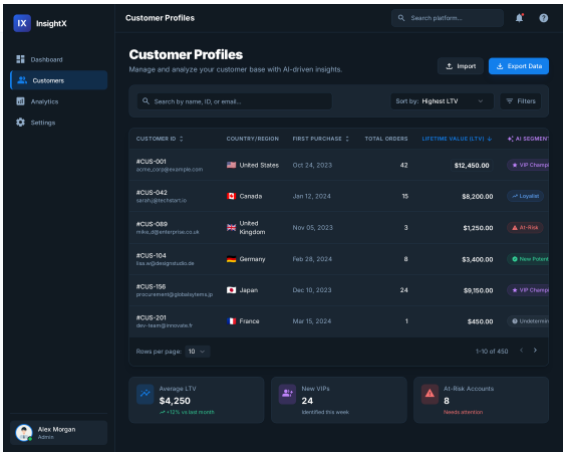
Product:



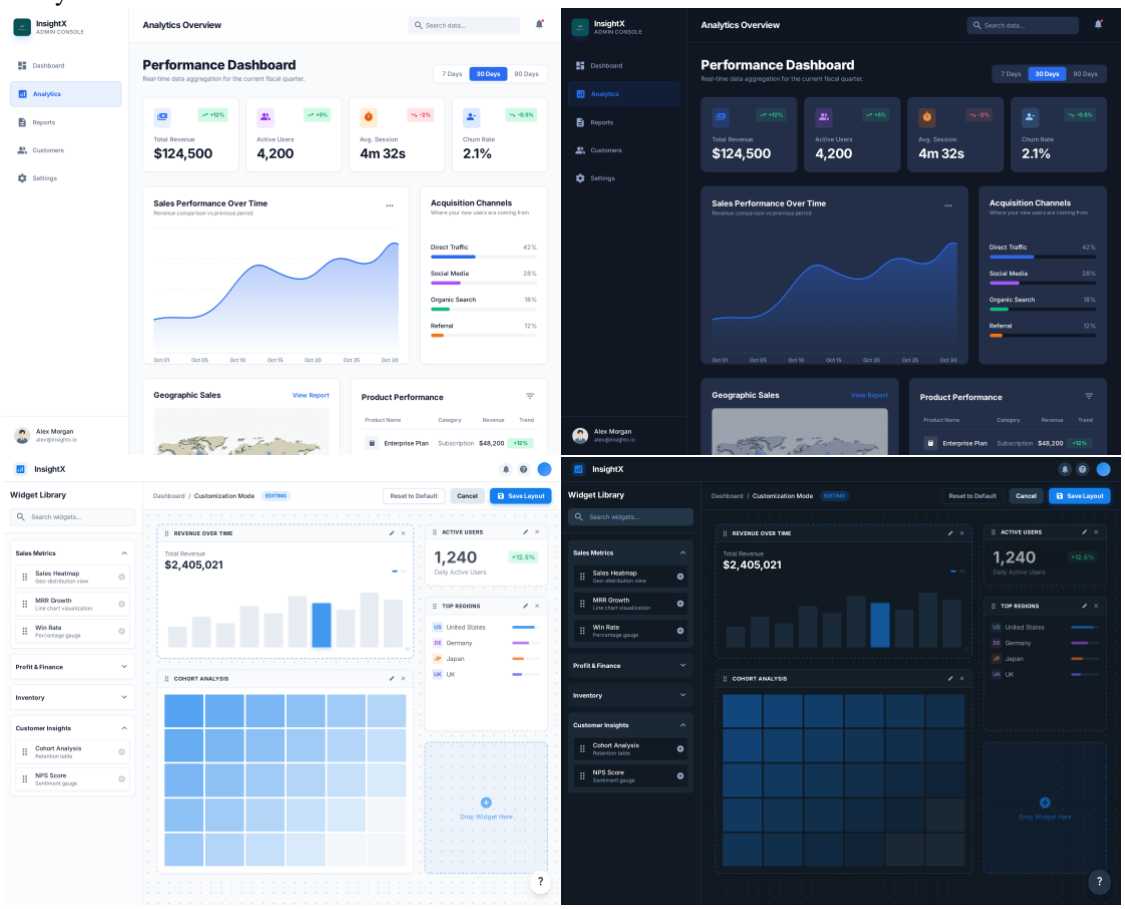


Customer :

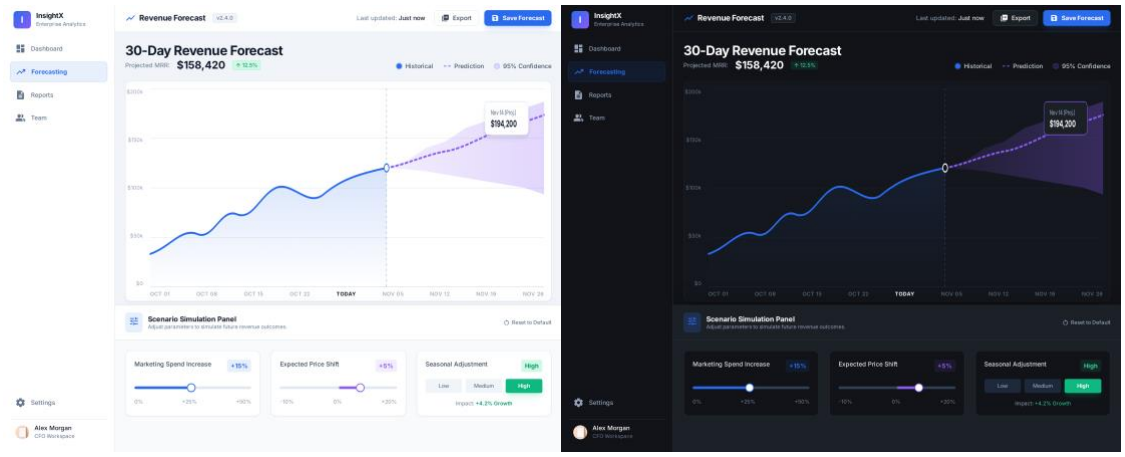




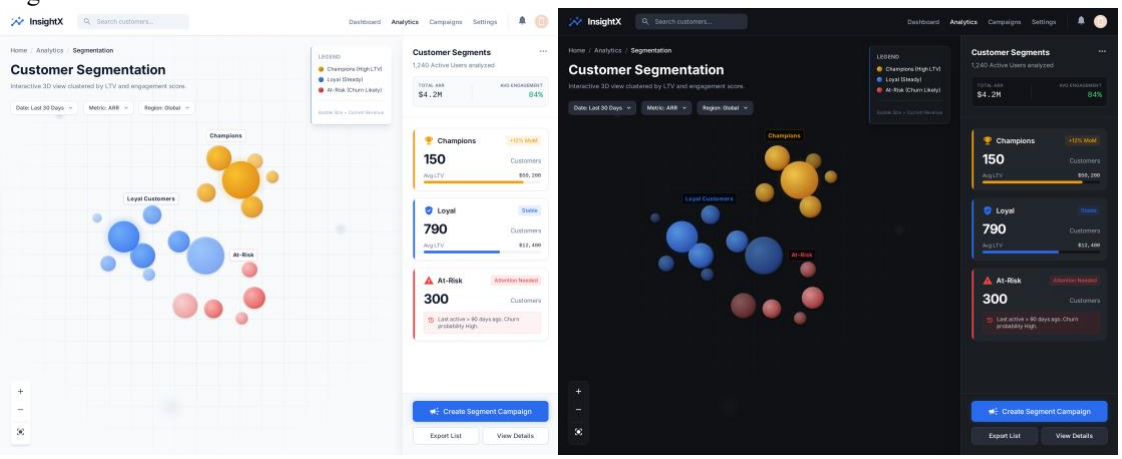
Analytics:



Forecasting:



Segmentation:



3.5 Feasibility Study

- **Technical:** The project is feasible using modern decoupled frameworks (FastAPI, React). The AI models (Prophet/BERT) are open-source and highly efficient for containerized environments.
- **Economic:** As a cloud-native SaaS model, initial infrastructure costs are low (AWS/Docker), with high potential for subscription-based revenue from e-commerce merchants.

3.6 Conclusion

The system design ensures a robust, scalable platform. The deliberate separation of concerns between the AI engine (Celery/Redis) and the Web interface (FastAPI/React) guarantees high performance and zero browser timeouts.

CHAPTER 4: Implementation

1.

1.1. Introduction

1.2. System implementation

1.3. Database design

1.4. User interfaces

1.5. Conclusion

CHAPTER 5: Conclusion and future work

2.

2.1. Conclusion

2.2. Future work

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